KERALA POLYTECHNIC · REVISION 2026

2041 | Semester II | Theory | 4 Credits | 60 Contact Hrs

Elements of Electrical & Electronics Engineering

Foundational principles of electricity, circuit laws, domestic installations, electronic devices, digital electronics, and recent trends — designed for Diploma students across multiple engineering disciplines.

Programme(s)	Biomedical Engg, Electronics & Communication, Electronics Engg, Electronics Engg (Embedded Systems), Electronics & Computer Engg, Integrated Circuit Design & Fabrication, Instrumentation Engg, Micro Electronics
Course Type	Theory (L:T:P:S = 4:0:0:0)
CIE / ESE	40 / 60
Self-Learning	90 hours total (6 hrs/week)

4
MODULES

OFFICIAL

Source Declaration

This handbook is derived from the official Kerala Polytechnic Revision 2026 syllabus for **Course 2041 — Elements of Electrical & Electronics Engineering**.

Document: 2041.pdf (pages 1–5) · **Revision:** REV2026 · **Generated:** 2026-07-30

📌 **Verified Inconsistencies in the Source PDF:**

- The pre-requisite table lists **2002A** (Physics) as I/II semester, **1002** (Mathematics) as I semester, and **1041** (Elementary Electronics) as I semester — but the course itself is Semester II. This is a typical articulation; the prerequisites are prior semesters.
- The CO–PO mapping table has total correlations of 11, 9, 2, etc. The per-row totals do not exactly match the sum of individual PO values in some rows (e.g., CO1: 3+3+2 = 8, but total shown as 11). This appears to be an artifact of the original table formatting; the handbook preserves the values as printed.
- Self-learning hours are listed as 6 hours per week for 15 weeks = 90 hours total, which is consistent.

All syllabus content, module topics, COs, and assessment rules are reproduced faithfully from the source. Educational elaboration (explanations, diagrams, worked examples, calculators) is provided for learning support and is clearly distinguished from official syllabus content.

AT A GLANCE

Course Dashboard

2041
COURSE CODE

II
SEMESTER

4
CREDITS

60
CONTACT HOURS

📖 Course Introduction

This course provides a foundational understanding of electrical and electronics engineering concepts. It covers the basic principles of electricity, circuit laws, and domestic electrical installations. The course also introduces the fundamentals of electronic devices, digital electronics, and recent trends in electronics. It aims to develop essential technical knowledge required for further studies and practical applications in electrical and electronics engineering.

📋 Course Objectives

1. To impart fundamental knowledge of electrical engineering principles and circuit laws for solving basic numerical and practical engineering problems.
2. To enable learners to identify and apply concepts of domestic electrical supply systems for analyzing electrical installations and estimating energy utilization.
3. To develop an understanding of static electricity, electrostatic discharge (ESD), ESD protection techniques, and the necessity of voltage regulation in electronic circuits.
4. To introduce the fundamentals of digital electronics, including number systems, number conversions, binary arithmetic, and basic logic gates for basic digital circuit analysis.

Teaching & Assessment Scheme

Teaching/Week	Self-Learning/Week	Total Hrs/Sem	Credits	CIE	ESE	Total
4:0:0:0 (L:T:P:S)	6	60 + 90 = 150	4	40	60	100

Course Pre-requisites

Topic	Course Code	Course Title	Semester
Basic knowledge in Physics	2002A	Engineering Physics for Applied Electrical Technology and Computing	I/II
Fundamental of Mathematics	1002	Fundamentals of Engineering Mathematics	I
Basic knowledge in Electronics	1041	Elementary Concepts of Electronics	I

GUIDE

How to Use This Handbook

1
Understand
Read each module's explanation, formulas, and worked examples.

2
Visualise
Study the SVG diagrams and circuit representations.

3
Apply
Use the interactive calculators and solve the practice problems.

4
Revise
Review the quick revision cards, Q&A, and model paper before exams.

🔍 Use the **Jump Menu** above to navigate quickly. Use **Print / PDF** to generate a printer-friendly version. Use the **Search** box at the bottom to find topics across the handbook.

COMPETENCIES

Course Outcomes

Course Outcomes (CO) with Bloom's Taxonomy Level

CO	Course Outcome	Bloom's Level
CO1	Demonstrate fundamental concepts of electrical engineering and circuit laws to solve numerical and practical engineering problems.	Apply
CO2	Identify and apply knowledge of domestic electrical supply system components to analyze installations and calculate energy utilisation.	Apply
CO3	Explain static electricity, electrostatic discharge, ESD precautions, and the importance of voltage regulation in electronic circuits.	Understand
CO4	Apply fundamentals of digital electronics including number systems, number conversions, binary arithmetic, and basic logic gates.	Apply

CO–PO Mapping (Course Articulation Matrix)

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	Total
CO1	3	3	2	–	–	–	–	–	–	–	–	11
CO2	3	2	2	2	–	–	–	–	–	–	–	9
CO3	2	2	–	–	2	2	–	–	–	–	1	2
CO4	3	2	–	–	2	1	–	–	–	–	1	2

Correlation: 1 = Low, 2 = Medium, 3 = High; – = No Correlation. Total shown as printed in source.

MAP

Curriculum Coverage

Module → Topics → Hours → CO

Module	Title	Major Topics	Hours	CO
I	Fundamental Laws in Electrical Engineering	Current, Voltage, Resistance; AC/DC; Ohm's Law; Law of Resistance; KCL/KVL; Voltage/Current Division; Faraday's Laws; AC Generation; Sinusoidal Waveform	15	CO1
II	Domestic Electrical Supply System and Energy Utilization	Basic terms; Supply system structure; Protection devices; Earthing; Safety; Wiring; Power factor; Electrical energy	15	CO2
III	Static Electricity, ESD, and Voltage Regulation	Static electricity; ESD; Precautions; Voltage regulation; Zener diode; Regulator ICs	15	CO3
IV	Digital Electronics Fundamentals	Number systems; Conversions; Binary arithmetic; Logic gates	15	CO4

CONNECTIONS

Module Roadmap

M1
Fundamental Laws
Circuit laws, Ohm, KCL/KVL, AC generation — the foundation.

M2
Domestic Supply
Distribution, protection, earthing, wiring, energy billing.

M3
ESD & Regulation
Static protection, Zener regulators, 78XX/79XX ICs.

M4
Digital Electronics
Number systems, binary arithmetic, logic gates — the digital foundation.

Modules progress from fundamental electrical laws → domestic applications → protection and regulation → digital electronics. Each module builds on the previous where relevant.

MODULE I

Fundamental Laws in Electrical Engineering

Introduction to current, voltage, resistance, AC/DC, Ohm's law, Kirchhoff's laws, Faraday's laws, AC generation, and sinusoidal waveforms.

🕐 15 hrs | 🎯 CO1 | 📊 Bloom: Apply

▶Introduction & Basic Electrical Quantities

Electricity is a fundamental form of energy resulting from the existence of charged particles (electrons and protons). The study of electrical engineering begins with understanding three primary quantities: **Current**, **Voltage**, and **Resistance**.

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Key Definitions:

Current (I) = Rate of flow of electric charge  →  I = Q / t
(units: Amperes, A)

Voltage (V) = Electrical potential difference between two points
  (units: Volts, V)

Resistance (R) = Opposition to current flow   (units: Ohms, Ω)
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Electric Current (I): The flow of electrons through a conductor. One ampere is one coulomb of charge passing a point per second.
Voltage (V): The "push" or pressure that drives current through a circuit. Also called electromotive force (EMF) or potential difference.
Resistance (R): The property of a material that opposes the flow of current. Depends on material, length, cross-sectional area, and temperature.

Malayalam support: Voltage □□□ "electrical pressure" □□□□□□□, Current charge flow rate □□□□, Resistance □ flow-□□ □□□□□□□ □□□□□□□□ □□□□□□□□□□□□□. □□□ analogy □□□□□□; exact definition formula-□□□□□□□.

💡 **Engineering Habit:** Always identify the three quantities (V, I, R) in any circuit problem. They are the foundation of all electrical analysis.

⚡ Ohm's Law Calculator

Voltage (V)

e.g. 12

Resistance (R)

e.g. 4

Enter values and click Calculate.

▶Concept of AC and DC

Direct Current (DC): Current that flows in one direction only. The magnitude may be constant or varying, but polarity does not change. Examples: batteries, solar cells, DC generators.
Alternating Current (AC): Current that periodically reverses direction. The magnitude varies sinusoidally with time. Examples: mains power supply (230V, 50Hz in India), AC generators.

Figure 1: Comparison of DC (constant) and AC (sinusoidal) waveforms.

Comparison: AC vs DC

Feature	AC	DC
Direction	Alternates periodically	Constant (one direction)
Magnitude	Varies sinusoidally	Constant or varying (non-reversing)
Generation	AC generators (alternators)	Batteries, solar cells, DC generators
Transmission	Efficient over long distances	Losses higher for long distances
Applications	Mains power, motors, transformers	Electronics, batteries, automotive

Malayalam support: AC □□□ current direction □□□□□□□□□ □□□□□□□□ (sinusoidal waveform), DC □□□ constant □□□ □□□ direction-□ □□□□□□□ flows.

▶Ohm's Law and Law of Resistance

Ohm's Law: The current through a conductor between two points is directly proportional to the voltage across the two points, provided the temperature remains constant.

$$V = I \times R \rightarrow I = V / R \rightarrow R = V / I$$

V = Voltage (V), I = Current (A), R = Resistance (Ω)

Law of Resistance: The resistance of a conductor is directly proportional to its length (L) and inversely proportional to its cross-sectional area (A), and depends on the material's resistivity (ρ).

$$R = \rho \times (L / A)$$

R = Resistance (Ω), ρ = Resistivity (Ω·m), L = Length (m), A = Cross-sectional area (m^2)

📐 **Worked Example — Ohm's Law**
Given: A 12V battery is connected to a 4Ω resistor.
Required: Current through the resistor.
Calculation: I = V / R = 12 / 4 = **3 A**
Answer: 3 amperes.

⚠️ **Common Mistake:** Ohm's Law applies only when temperature is constant. For non-ohmic materials (diodes, thermistors), V–I relationship is non-linear.

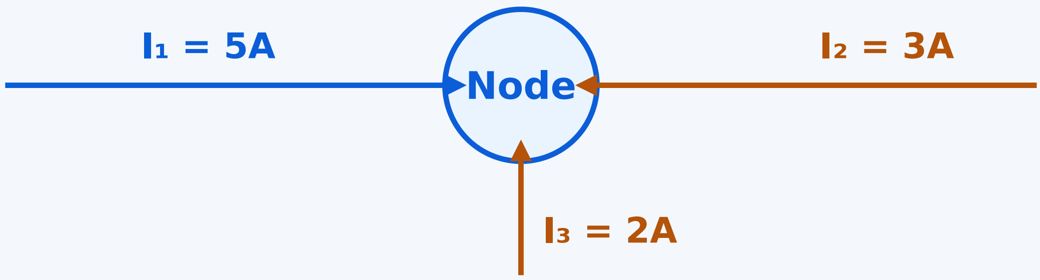
▶Kirchhoff's Laws (KCL and KVL)

Kirchhoff's Current Law (KCL): The algebraic sum of currents entering a node (junction) is zero. Equivalently, total current entering = total current leaving.

$$\Sigma I_{in} = \Sigma I_{out}$$

Kirchhoff's Voltage Law (KVL): The algebraic sum of all voltages around any closed loop in a circuit is zero.

$$\Sigma V = 0 \quad \text{(around any closed loop)}$$



KCL: $I_1 = I_2 + I_3 \rightarrow 5 = 3 + 2$ ✓

Figure 2: KCL at a node — total current entering equals total current leaving.

📐 **Worked Example — KVL**

Given: A series circuit with $V_S = 10V$, $R_1 = 2\Omega$, $R_2 = 3\Omega$.

Required: Voltages across R_1 and R_2.

Calculation: $I = V / (R_1+R_2) = 10/5 = 2A$. $V_1 = I\times R_1 = 4V$, $V_2 = I\times R_2 = 6V$.

KVL check: 10 - 4 - 6 = 0 ✓

Answer: $V_1 = 4V$, $V_2 = 6V$.

▶Voltage Division and Current Division Rules

Voltage Division Rule (VDR): In a series circuit, the voltage across a resistor is proportional to its resistance.

$$V_x = V_{total} \times (R_x / R_{total})$$

V_x = Voltage across resistor R_x, R_{total} = sum of all series resistances

Current Division Rule (CDR): In a parallel circuit, the current through a branch is inversely proportional to its resistance.

$$I_x = I_{total} \times (R_{total} / R_x) \quad \text{where } R_{total} = (R_1 \times R_2)/(R_1+R_2) \text{ for two resistors}$$

I_x = Current through branch with resistance R_x

📐 **Worked Example — Voltage Division**

Given: V_s = 24V, R_1 = 6Ω, R_2 = 2Ω in series.

Required: Voltage across R_2.

Calculation: $V_2 = 24 \times (2 / (6+2)) = 24 \times 0.25 =$ **6V**

Answer: 6 volts across R_2.

▸Faraday's Laws of Electromagnetic Induction

Faraday's First Law: Whenever the magnetic flux linked with a coil changes, an EMF (voltage) is induced in the coil. The magnitude of the induced EMF is proportional to the rate of change of flux linkage.
Faraday's Second Law: The magnitude of the induced EMF is directly proportional to the rate of change of magnetic flux linkage.

$$E = -N \times (d\Phi / dt)$$

E = Induced EMF (V), N = Number of turns, Φ = Magnetic flux (Wb), dΦ/dt = Rate of change of flux

Lenz's Law: The direction of the induced current is such that it opposes the change that produced it (the negative sign in the formula).
Fleming's Right-Hand Rule: Used to determine the direction of induced current in a conductor moving in a magnetic field. Thumb = Motion, Forefinger = Field, Middle finger = Current.

💡 **Application:** Faraday's laws are the basis for generators, transformers, and induction motors.

▶Generation of AC Voltage and Sinusoidal Waveforms

AC Voltage Generation: When a coil rotates in a uniform magnetic field, the flux linkage changes sinusoidally, producing a sinusoidal EMF. This is the principle of the AC generator (alternator).
Sinusoidal Waveform: The standard AC waveform can be described by:

$$v(t) = V_m \times \sin(\omega t + \varphi)$$

$v(t)$ = Instantaneous voltage, V_m = Peak voltage, ω = Angular frequency ($2\pi f$), φ = Phase angle

Key parameters: Peak value (V_m), RMS value ($V_{rms} = V_m/\sqrt{2}$), Average value, Frequency (f), Time period ($T = 1/f$).

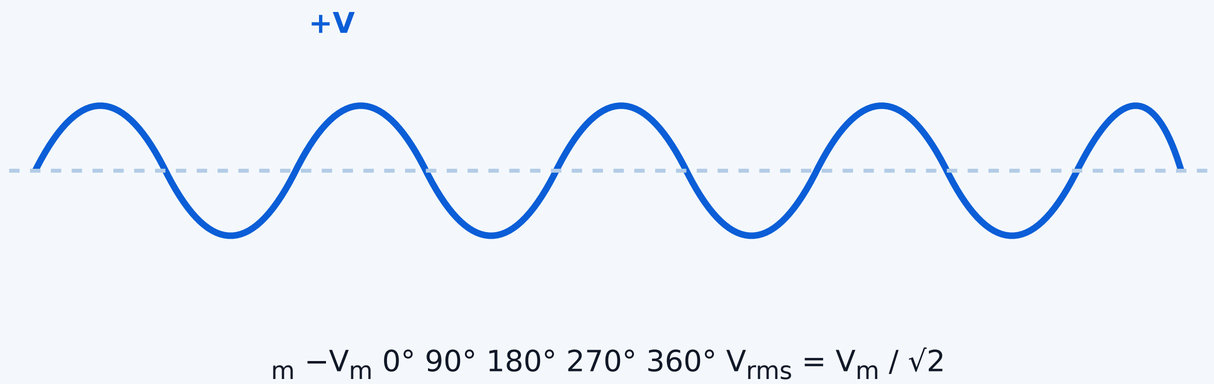


Figure 3: Sinusoidal AC waveform showing one complete cycle, peak values, and RMS level.

Malayalam support: AC voltage generation — coil magnetic field-□ rotate □□□□□□□□□□□ sinusoidal waveform □□□□□□□□□□□. V_{rms} □□□ effective value.

🧠 **Module I — Quick Summary:** Ohm's Law (V=IR), KCL (current in = out), KVL (loop voltages sum to zero), voltage/current division rules, Faraday's induction laws, and AC sinusoidal representation form the core of electrical circuit analysis.

MODULE II

Domestic Electrical Supply System and Energy Utilization

Basic terms, supply system structure, protection devices, earthing, safety, wiring, power factor, and electrical energy billing.

🕐 15 hrs 🎯 CO2 📊 Bloom: Apply

▶Basic Terms in Electrical Circuits

Circuit: A closed path through which electric current flows.

Load: Any device that consumes electrical energy (e.g., bulb, motor, heater).

Phase (Live): The conductor that carries current at a high potential (typically 230V in India).

Neutral: The return path conductor, at or near earth potential.

Earth (Ground): A safety connection to the earth, providing a low-resistance path for fault currents.

Single-phase supply: Two wires — Phase and Neutral (230V, 50Hz in India). Used for residential lighting and small appliances.

Three-phase supply: Four wires — Three phases (R, Y, B) and Neutral (415V line-to-line, 230V line-to-neutral). Used for industrial motors and heavy loads.

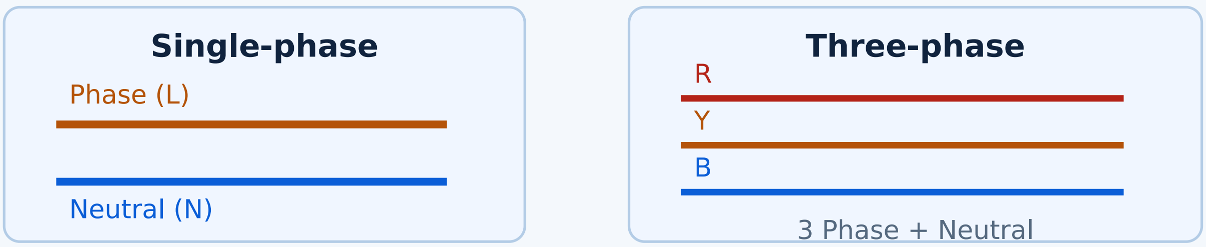


Figure 4: Single-phase (L+N) vs Three-phase (R+Y+B+N) supply systems.

> **Malayalam support:** Single-phase — domestic □□□□□□ (230V), Three-phase — industrial (415V). Phase □□□ live wire; Neutral return path.

▶Structure of a Domestic Electrical Supply System

A typical domestic supply system consists of:

1. **Service Connection:** The overhead or underground cable from the distribution pole to the consumer's premises.
2. **Energy Meter:** Measures the electrical energy consumed (in kWh).
3. **Main Switch (Isolator):** A switch that disconnects the entire supply to the premises for safety.
4. **Distribution Board (DB):** Contains MCBs, fuses, and neutral links; distributes supply to different circuits.
5. **Final Sub-circuits:** Lighting, power, and appliance circuits.

> 💡 **Engineering Habit:** Always identify the main switch and energy meter location in any domestic installation — they are the entry points for supply and safety.

▶Protection Devices — Fuses, MCB, ELCB, Neutral Link

Fuse: A sacrificial device that melts when current exceeds a safe value, opening the circuit. Types: *Reversible (rewirable)* and *Cartridge (HRC)*.
Miniature Circuit Breaker (MCB): A protective switch that trips automatically on overcurrent or short-circuit. Can be reset manually. Common ratings: 6A, 10A, 16A, 20A, 32A.
Earth Leakage Circuit Breaker (ELCB) / Residual Current Device (RCD): Detects leakage currents (imbalance between phase and neutral) and trips to prevent electric shock.
Neutral Link: A connection point for all neutral wires in the distribution board, providing a common return path.

Protection Devices Comparison

Device	Protection Against	Reset	Common Rating
Fuse (rewirable)	Overcurrent, short-circuit	Replace wire	5A, 15A, 30A
Fuse (cartridge/HRC)	Overcurrent, short-circuit	Replace cartridge	2A–100A
MCB	Overcurrent, short-circuit	Manual reset	6A, 10A, 16A, 20A, 32A
ELCB / RCD	Earth leakage (shock)	Manual reset	30mA, 100mA, 300mA

> ⚠️ **Safety Note:** Never bypass a fuse or MCB. Use the correct rating for each circuit. ELCBs are mandatory for shock protection in modern installations.

▶Earthing — Need, Types, and Materials

Need for Earthing: To provide a safe path for fault currents to dissipate into the earth, protecting humans and equipment from electric shock and damage.

Types of Earthing:

- **Plate Earthing:** A copper or GI plate buried in the earth (typically 2.4m × 1.2m × 3mm).
- **Pipe Earthing:** A galvanized iron (GI) pipe (typically 38mm diameter, 2.5m length) buried vertically in the earth.

Earthing Materials:

- **Earth Wire:** Copper or GI wire connecting the equipment to the earth electrode.
- **Earth Electrode:** The conductive plate, pipe, or rod buried in the earth.

> 💡 **Application:** All electrical installations must have a proper earthing system. Resistance to earth should be less than 5Ω for domestic systems.

▶Electrical Safety and Electrical Shock

Causes of Electric Shock: Direct contact with live conductors, faulty insulation, wet conditions, and improper earthing.

Effects of Electric Shock: Depends on current magnitude, path, and duration. Effects range from mild tingling to ventricular fibrillation (cardiac arrest) and burns.

Prevention Measures:

- Use insulated tools and wear rubber-soled shoes.
- Install ELCBs/RCDs for shock protection.
- Ensure proper earthing of all equipment.
- Keep electrical panels dry and accessible.
- Never touch electrical equipment with wet hands.
- Use appropriate PPE (Personal Protective Equipment).

> ⚠️ **Safety Note:** Current as low as 10mA can be dangerous. 50mA can cause ventricular fibrillation. Always treat electrical systems with respect.

▶Wiring in Residential Buildings

Types of Wiring Systems:

- **Cleat Wiring:** Wires supported on porcelain or wooden cleats. Outdated, used temporarily.
- **Batten Wiring:** Wires laid on wooden battens with porcelain clips. Used in older installations.
- **Casing and Capping Wiring:** Wires enclosed in wooden or plastic casing with a capping cover. Common in older buildings.
- **Conduit Wiring:** Wires drawn through metal or PVC pipes. *Open conduit* is surface-mounted; *concealed conduit* is embedded in walls/ceilings. This is the most common modern system.

Wiring Materials: PVC-insulated copper/aluminum wires, PVC conduit pipes, junction boxes, switch boxes, and accessories.

> 💡 **Engineering Habit:** Conduit wiring (concealed) is preferred for residential buildings due to safety, aesthetics, and durability.

▶Power Factor and Types of Powers

Power Factor (PF): The ratio of real power (active power) to apparent power. PF = cos φ, where φ is the phase angle between voltage and current.

$$P = V \times I \times \cos\varphi \quad \text{(for single-phase AC)}$$

P = Active/Real Power (W), V = Voltage (V), I = Current (A), cos φ = Power Factor

Types of Powers:

- **Active Power (P):** Actual power consumed by the load, measured in Watts (W).
- **Reactive Power (Q):** Power exchanged between source and load (inductive/capacitive), measured in VAR (Volt-Ampere Reactive).
- **Apparent Power (S):** The product of V and I, measured in VA (Volt-Ampere). $S = \sqrt{P^2 + Q^2}$.

📐 **Worked Example — Power Factor**

Given: A motor draws 5A at 230V with PF = 0.8.

Required: Active power, reactive power, apparent power.

Calculation: $S = 230 \times 5 = 1150$ VA. $P = 1150 \times 0.8 = 920$ W. $Q = \sqrt{1150^2 - 920^2} = \sqrt{1{,}322{,}500 - 846{,}400} = \sqrt{476{,}100} \approx 690$ VAR.

Answer: P = 920W, Q ≈ 690 VAR, S = 1150 VA.

Malayalam support: Power factor — active power (□□□□□□□□□□□□□□ power) and apparent power (total power) □□□□□□□□□□□ ratio. PF □□□□□□□□□ losses □□□□□.

▸Electrical Energy — Units and Billing

Electrical Energy: The total work done by electrical power over time.

$$E = P \times t$$

E = Energy (Joules or kWh), P = Power (W), t = Time (s or h)

Commercial Unit: Kilowatt-hour (kWh). 1 kWh = 1000 W × 1 hour = 3.6×10^6 Joules.

Energy Consumption Calculation: Total energy (kWh) = (Power rating in W × Hours of use per day × Number of days) / 1000.

📐 **Worked Example — Electricity Bill**

Given: A home has: 5 bulbs (60W each) used 6 hrs/day, 1 fan (80W) used 10 hrs/day, 1 TV (100W) used 4 hrs/day. Tariff = ₹5 per kWh.

Required: Monthly energy consumption and bill (30 days).

Calculation: Bulbs: 5×60×6×30 = 54,000 Wh = 54 kWh. Fan: 80×10×30 = 24,000 Wh = 24 kWh. TV: 100×4×30 = 12,000 Wh = 12 kWh. Total = 54+24+12 = 90 kWh. Bill = 90 × ₹5 = **₹450**.

Answer: 90 kWh, ₹450 per month.

💡 **Energy & Bill Estimator**

Power (W): e.g. 100

Hours per day: e.g. 5

Days: e.g. 30

Tariff (₹/kWh): e.g. 5

Enter values and click Calculate.

🧠 **Module II — Quick Summary:** Domestic supply uses single-phase (230V) with phase, neutral, earth. Protection devices (fuses, MCB, ELCB) are critical for safety. Earthing provides a safe fault path. Power factor affects efficiency; energy billing is based on kWh consumption.

MODULE III

Static Electricity, ESD, and Voltage Regulation

Static electricity, electrostatic discharge (ESD), ESD precautions, voltage regulation, Zener diode, and regulator ICs.

🕐 15 hrs | 🎯 CO3 | 📊 Bloom: Understand

▶Static Electricity and Electrostatic Discharge (ESD)

Static Electricity: The buildup of electric charge on the surface of an object. It occurs when two materials are rubbed together, transferring electrons from one to the other. The charge remains stationary (static) until it is discharged.

Electrostatic Discharge (ESD): The sudden flow of electricity between two objects at different electrical potentials. ESD can occur when a charged object comes near or contacts a conductive object.

Impact of ESD on Electronic Components: ESD can cause:

- Catastrophic failure (immediate destruction of the component).
- Latent failure (degradation that leads to failure later).
- Disruption of logic states in digital circuits.

⚠️ **Common Mistake:** Many people think static electricity is harmless because they don't feel small discharges. However, even discharges as low as 10V can damage sensitive electronic components (CMOS ICs, MOSFETs).

Malayalam support: Static charge □□□□□ materials □□□□□□ friction □□□□□□□□□□□□ build-up □□□□□□□. ESD □□□□□□□□□□□□ components-□□ damage □□□□□□□□□□.

▶ESD Precautions

ESD Prevention Techniques:

- **Grounding:** Use wrist straps and heel straps to ground personnel.
- **Antistatic Workstations:** Use conductive mats and work surfaces.
- **Antistatic Packaging:** Store and transport components in antistatic bags (pink or black conductive bags).
- **Humidity Control:** Maintain 40–60% relative humidity to reduce static buildup.
- **Ionizers:** Use air ionizers to neutralize static charges in the environment.
- **Proper Handling:** Handle components by their edges; avoid touching pins or leads.

⚠️ **Safety Note:** In a laboratory or manufacturing environment, always wear a grounded wrist strap before handling sensitive electronic components. Never work on live circuits.

▶Need for Voltage Regulation in Electronic Circuits

Why Voltage Regulation is Needed:

- **Stable Operation:** Electronic circuits (digital ICs, microcontrollers, amplifiers) require a stable supply voltage to function correctly.
- **Load Variations:** When load current changes, the voltage tends to drop. Regulation keeps the output voltage constant.
- **Input Voltage Fluctuations:** Mains voltage can vary (e.g., 230V ± 10%). Regulation compensates for these variations.
- **Noise Suppression:** Voltage regulators help filter out ripple and noise from the power supply.

Regulation Parameters:

- **Load Regulation:** Change in output voltage when load current changes (from no-load to full-load).
- **Line Regulation:** Change in output voltage when input voltage changes.

$$\text{Load Regulation (\%)} = (V_{NL} - V_{FL}) / V_{FL} \times 100$$

V_{NL} = No-load voltage, V_{FL} = Full-load voltage

▸Zener Diode as a Voltage Regulator

Zener Diode: A special type of diode designed to operate in the reverse breakdown region. When reverse-biased and the voltage exceeds the Zener voltage (V_Z), the diode conducts and maintains a nearly constant voltage across it.

Working Principle: In a simple Zener regulator circuit, a series resistor (R_s) limits the current, and the Zener diode shunts excess current to maintain a constant output voltage across the load.

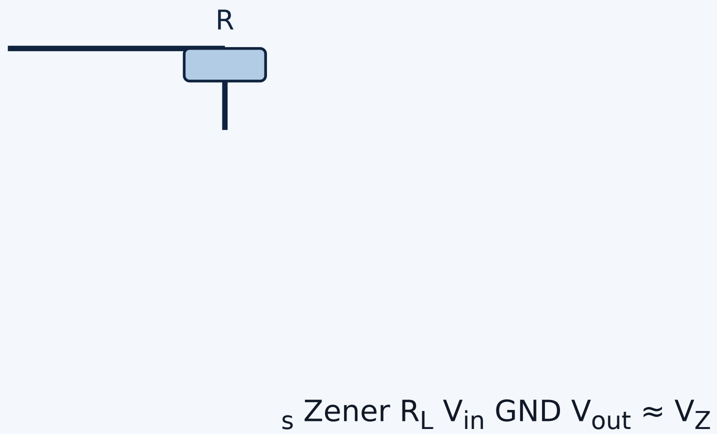


Figure 5: Zener diode voltage regulator circuit. Output voltage is clamped to V_Z.

Advantages: Simple, low cost, and effective for low-current applications.

Disadvantages: Not efficient for high currents; power dissipation in the Zener can be high.

> **Malayalam support:** Zener diode reverse breakdown region-□ operate □□□□□□□□□□; output voltage V_Z-□ constant □□□ □□□□□□□□□□□□□□.

▶Voltage Regulator ICs — 78XX, 79XX, and LM317

Fixed Voltage Regulator ICs — 78XX and 79XX Series:

- **78XX Series:** Positive voltage regulators. The last two digits indicate the output voltage (e.g., 7805 = +5V, 7812 = +12V, 7824 = +24V).
- **79XX Series:** Negative voltage regulators. (e.g., 7905 = −5V, 7912 = −12V).

Typical Connection: Input voltage → 78XX IC → output voltage. Requires input voltage at least 2–3V higher than the desired output (dropout voltage). Bypass capacitors on input and output are recommended for stability.

Variable Voltage Regulator — LM317:

- Adjustable positive voltage regulator (1.25V to 37V).
- Uses two external resistors (R_1, R_2) to set the output voltage.
- Output voltage: $V_{out} = 1.25 \times (1 + R_2/R_1) + I_{adj} \times R_2$
- Common applications: adjustable power supplies, bench power supplies.

Common 78XX / 79XX Regulator ICs

IC	Output Voltage	Type	Typical Application
7805	+5V	Positive	Digital logic, microcontrollers
7809	+9V	Positive	Analog circuits, op-amps
7812	+12V	Positive	Motor drivers, audio amps
7815	+15V	Positive	Instrumentation, op-amps
7824	+24V	Positive	Industrial control, relays
7905	−5V	Negative	Dual-supply op-amp circuits
7912	−12V	Negative	Audio, instrumentation

💡 **Engineering Habit:** Always check the datasheet for the specific regulator IC. Pay attention to maximum input voltage, current rating, and heat sinking requirements.

🧠 **Module III — Quick Summary:** Static charge buildup leads to ESD, which can damage electronics. ESD precautions include grounding, antistatic packaging, and proper handling. Voltage regulation is essential for stable circuit operation; Zener diodes and IC regulators (78XX, 79XX, LM317) are common solutions.

MODULE IV

Digital Electronics Fundamentals

Introduction to digital electronics, number systems, number conversions, binary arithmetic, and logic gates.

🕐 15 hrs | 🎯 CO4 | 📊 Bloom: Apply

▶Introduction to Digital Electronics

Digital Electronics: The branch of electronics that deals with discrete (digital) signals — signals that have only two states: HIGH (1) and LOW (0).

Advantages of Digital Systems:

- Noise immunity — digital signals can be regenerated without degradation.
- Accurate and reproducible results.
- Easier to store and process information.
- Flexibility — digital systems can be programmed (microprocessors, FPGAs).

Applications: Computers, smartphones, digital communication, control systems, medical devices, consumer electronics.

Analog vs Digital Systems

Feature	Analog	Digital
Signal type	Continuous (any value)	Discrete (0 or 1)
Noise immunity	Low — noise affects the signal directly	High — noise needs to exceed threshold
Accuracy	Limited by component tolerances	High — determined by number of bits
Processing	Hardware-dependent	Flexible — programmable
Examples	Audio amplifiers, thermometers	Computers, digital watches

> **Malayalam support:** Digital electronics — signals 0 (LOW) and 1 (HIGH) □□□ only □□□□□ states. Analog-□□□□□□□ noise immunity □□□□□□□□□.

▶Number Systems — Decimal, Binary, and Hexadecimal

Number Systems:

- **Decimal (Base-10):** Uses digits 0–9. Example: $(1234)_{10} = 1\times10^3 + 2\times10^2 + 3\times10^1 + 4\times10^0$
- **Binary (Base-2):** Uses digits 0 and 1. Example: $(1011)_2 = 1\times2^3 + 0\times2^2 + 1\times2^1 + 1\times2^0 = 11_{10}$
- **Hexadecimal (Base-16):** Uses digits 0–9 and letters A–F (A=10, B=11, C=12, D=13, E=14, F=15). Example: $(1A)_{16} = 1\times16^1 + 10\times16^0 = 26_{10}$

$$\text{General: } (N)_b = d_{n-1}\times b^{n-1} + \ldots + d_0\times b^0$$

b = base, d_i = digits, n = number of digits

Number System Comparison

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

▶Conversion of Numbers

Binary to Decimal: Multiply each bit by $2^{position}$ and sum.

Decimal to Binary: Repeatedly divide by 2, noting remainders (read bottom to top).

Hexadecimal to Decimal: Multiply each digit by $16^{position}$ and sum.

Decimal to Hexadecimal: Repeatedly divide by 16, noting remainders (read bottom to top).

Binary to Hexadecimal: Group binary digits into sets of 4 from the right; convert each group to hex.

Hexadecimal to Binary: Convert each hex digit to its 4-bit binary equivalent.

📐 **Worked Example — Binary to Decimal**

Given: Binary number $(1101)_2$

Calculation: $1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0 = 8 + 4 + 0 + 1 = 13$

Answer: $(1101)_2 = (13)_{10}$

📐 **Worked Example — Decimal to Binary**

Given: Decimal number $(25)_{10}$

Calculation: $25 \div 2 = 12$ rem 1; $12 \div 2 = 6$ rem 0; $6 \div 2 = 3$ rem 0; $3 \div 2 = 1$ rem 1; $1 \div 2 = 0$ rem 1 → read bottom up: 11001

Answer: $(25)_{10} = (11001)_2$

🔢 **Number Base Converter**

Enter number

e.g. 1010 or 1A

From base

Decimal (10)

To base

Decimal (10)

Enter a number and select bases.

▶Binary Arithmetic — 1's Complement, 2's Complement, Addition, Subtraction

1's Complement: Flip all bits (0→1, 1→0).

2's Complement: 1's complement + 1. Used for representing negative numbers in binary.

Binary Addition: 0+0=0, 0+1=1, 1+0=1, 1+1=10 (carry 1).

Binary Subtraction: Can be performed using 2's complement: A − B = A + (2's complement of B).

📐 **Worked Example — Binary Addition**

Given: $(1011)_2 + (0110)_2$

Calculation: 1011 + 0110 = 10001 (with carry)

Answer: $(1011)_2 + (0110)_2 = (10001)_2 = (17)_{10}$

📐 **Worked Example — Binary Subtraction using 2's Complement**

Given: $(1001)_2 - (0011)_2$

Calculation: 2's complement of 0011: 1100 + 1 = 1101. 1001 + 1101 = 10110 → discard carry → 0110

Answer: $(1001)_2 - (0011)_2 = (0110)_2 = (6)_{10}$

Malayalam support: 2's complement negative numbers-□□ represent □□□□□□□ □□□□□□□□□□□□□□. Binary subtraction 2's complement □□□□□□□□□□ addition □□□ convert □□□□□□□.

▶Logic Gates — NOT, AND, OR, NAND, NOR, ExOR, ExNOR

Logic gates are the basic building blocks of digital circuits. Each gate has a truth table, a logic symbol, and a Boolean expression.

Logic Gates — Symbols, Expressions, and Truth Tables

Gate	Symbol	Expression	Truth Table
NOT	▷ with bubble	$Y = \bar{A}$	0→1, 1→0
AND	▷ shape	$Y = A \cdot B$	1 only when both inputs are 1
OR	▷ shape (curved)	$Y = A + B$	1 when at least one input is 1
NAND	AND + bubble	$Y = \overline{(A \cdot B)}$	0 only when both inputs are 1
NOR	OR + bubble	$Y = \overline{(A + B)}$	1 only when both inputs are 0
ExOR	▷ with extra curve	$Y = A \oplus B$	1 when inputs are different
ExNOR	ExOR + bubble	$Y = \overline{(A \oplus B)}$	1 when inputs are same

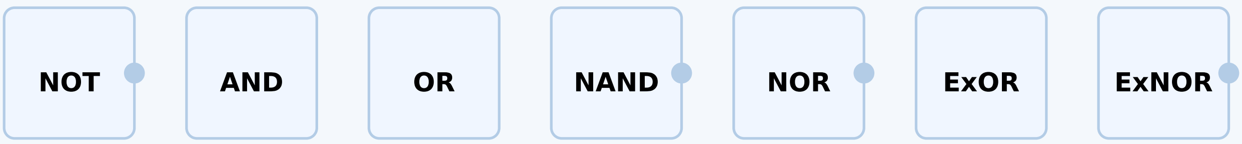


Figure 6: Standard symbols for the seven basic logic gates.

Applications:

- NOT: Invert signals, oscillators.
- AND: Enable/disable signals, digital comparators.
- OR: Signal merging, priority encoders.
- NAND, NOR: Universal gates — can build any other gate.
- ExOR, ExNOR: Parity generators, comparators, adders.

> 💡 **Engineering Habit:** NAND and NOR are called "universal gates" because any digital circuit can be built using only these gates. This is useful for IC design.

> 🧠 **Module IV — Quick Summary:** Digital electronics uses binary (0/1) signals. Number systems (binary, decimal, hex) are interconvertible. Binary arithmetic uses

1's and 2's complement for subtraction. Logic gates (NOT, AND, OR, NAND, NOR, ExOR, ExNOR) are the building blocks of all digital circuits.

REFERENCE

Formula Bank

$V = I \cdot R$

Ohm's Law (Module I)

$R = \rho \cdot L / A$

Law of Resistance (Module I)

$\Sigma I_{in} = \Sigma I_{out}$

KCL — Node current (Module I)

$\Sigma V = 0$

KVL — Loop voltage (Module I)

$V_X = V_T \cdot (R_X/R_T)$

Voltage Division (Module I)

$I_X = I_T \cdot (R_T/R_X)$

Current Division (Module I)

$E = -N \cdot d\Phi/dt$

Faraday's Law (Module I)

$v(t) = V_m \cdot \sin(\omega t + \varphi)$

AC Sinusoidal (Module I)

$P = V \cdot I \cdot \cos \varphi$

Active Power (Module II)

$S = V \cdot I$

Apparent Power (Module II)

$E = P \cdot t$

Electrical Energy (Module II)

$Reg\ (\%) = (V_{NL} - V_{FL}) / V_{FL} \times 100$

Load Regulation (Module III)

$V_{out} \approx V_Z$

Zener Regulation (Module III)

$(N)_b = \Sigma\ d_i \cdot b^i$

Number system expansion (Module IV)

A − B = A + (2's comp of B)

Binary subtraction (Module IV)

$Y = A \cdot B,\ Y = A + B,\ Y = \bar{A}$, etc.

Logic gate expressions (Module IV)

VISUALS

Diagram Library for Rapid Revision

📊 AC vs DC Waveforms

Comparison of sinusoidal AC and constant DC.

View in Module I →

🔀 KCL Node Diagram

Illustrates current entering and leaving a node.

View in Module I →

⚡ Single/Three-phase Supply

Visual comparison of domestic and industrial supply.

View in Module II →

🔌 Zener Regulator Circuit

Zener diode used as a voltage regulator.

View in Module III →

💠 Logic Gate Symbols

Standard symbols for all seven basic gates.

View in Module IV →

📈 Sinusoidal Waveform

Peak, RMS, and cycle parameters of AC.

View in Module I →

SELF-LEARNING

Suggested Student Activities

Official activities from the syllabus — 6 hours per week, 15 weeks total.

WEEK 1–4 · MODULE I

1. Prepare a list of electricity generation methods used in India. Create a chart of generating stations with generating power.

2. Study the difference between renewable and non-renewable energy sources.

3. Observe household appliances — list whether they use AC or DC, and identify power ratings.

4. Prepare a document on major stakeholders in Kerala's electricity sector.

WEEK 5–7 · MODULE II

5. Identify the type of wiring system used in your home and classroom. Write two reasons why this wiring method may have been selected.

6. Walk around your home and list the electrical safety measures used. Write how each measure helps in preventing accidents.

7. Prepare a list of major electrical equipment used in your home. Identify power ratings and estimate the monthly electricity bill. Compare with the actual bill.

WEEK 8–11 · MODULE III

8. Prepare a comparison chart between ordinary (Transformer type) DC power supply with SMPS.

9. Download the data sheets of 78XX series ICs. Prepare a chart showing its characteristics.

10. Study the internal diagram of LM317 and list out the typical applications of this IC.

11. Study the troubleshooting of ordinary power supply using a video tutorial. List the troubleshooting steps.

WEEK 12–15 · MODULE IV

12. Download the data sheets of some ICs used as two input basic logic gates. List the characteristics.

13. Explore the major laws in the field of digital electronics.

14. Prepare a list of digital devices that are used in your day to day life.

15. Prepare a document on the historical milestones in the field of Digital electronics.

EXAMS

Assessment Methodology

Assessment Scheme — CIE and ESE

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
Component	Attendance	Self-learning (M1)	Self-learning (M2)	Self-learning (M3)	Self-learning (M4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration	–	–	–	–	–	2 hrs	2 hrs	2.5 hrs
Exam Mark	–	–	–	–	–	15	15	50
Converted to	5	–	–	–	–	15	15	50
Total Marks	40 CIE			60 ESE			100 Total	
Tentative Schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern — Theory

Series Exam (2 hours, 25 marks)

Time	Module	Part A Marks	Part B Marks	Part C (with choice) Marks	Total
2 hours	Any one module	2 × 1 = 2	1 × 3 = 3	1 × 3 = 3	8
	Any one module	2 × 1 = 2	1 × 3 = 3	1 × 3 = 3	8
Total		4	6	9	**25**

End Semester Examination Pattern — Theory

ESE (2.5 hours, 60 marks)

Module	Part A (2 Qs from each module, 1 mark each)		Part B (2 out of 3 Qs from each module, 3 marks each)		Part C (1 set question with choice from each module, 7 marks each)		Total
1	2	2	2	6	1	7	15
2	2	2	2	6	1	7	15
3	2	2	2	6	1	7	15
4	2	2	2	6	1	7	15
Total	**8**	**8**	**8**	**24**	**4**	**28**	**60**

PRACTICE

Module-wise Questions with Answers

Module I — Fundamental Laws

Q: State and explain Ohm's Law. What are its limitations?

A: Ohm's Law states that the current through a conductor between two points is directly proportional to the voltage across the two points, provided the temperature remains constant. $V = I \times R$. Limitations: (1) It does not apply to non-ohmic devices (diodes, transistors, thermistors). (2) It is valid only for constant temperature. (3) It does not apply to gases or electrolytes where current is not directly proportional to voltage.

Q: State Kirchhoff's Current Law and Voltage Law. Give a numerical example for each.

A: **KCL:** The algebraic sum of currents entering a node is zero (or total current entering = total current leaving). **KVL:** The algebraic sum of voltages around any closed loop is zero. **Example (KCL):** At a node, $I_1 = 5A$ entering, $I_2 = 3A$ and $I_3 = 2A$ leaving → $5 = 3 + 2$ ✓. **Example (KVL):** For a series circuit with $V_s = 10V$, $R_1 = 2\Omega$, $R_2 = 3\Omega$: $I = 2A$, $V_1 = 4V$, $V_2 = 6V$ → $10 - 4 - 6 = 0$ ✓.

Q: Explain Faraday's laws of electromagnetic induction. State Lenz's Law.

A: **Faraday's First Law:** Whenever the magnetic flux linked with a coil changes, an EMF is induced. **Faraday's Second Law:** The magnitude of the induced EMF is proportional to the rate of change of flux linkage. $E = -N\cdot(d\Phi/dt)$. **Lenz's Law:** The direction of the induced current is such that it opposes the change that produced it (the negative sign in the equation).

Module II — Domestic Supply & Energy

Q: Draw the structure of a domestic electrical supply system. Explain the function of each part.

A: A domestic supply system consists of: (1) **Service Connection** — brings supply from distribution pole to the premises. (2) **Energy Meter** — measures consumption in kWh. (3) **Main Switch** — disconnects the entire supply for safety. (4) **Distribution Board** — contains MCBs, fuses, neutral link; distributes to sub-circuits. (5) **Final Sub-circuits** — supply lighting, power, and appliances.

Q: What is the difference between a fuse and an MCB? List the types of protection devices used in domestic installations.

A: **Fuse:** A sacrificial device that melts on overcurrent, needs replacement. **MCB:** A protective switch that trips on overcurrent/short-circuit, can be reset manually. **Protection devices:** Rewirable fuse, Cartridge (HRC) fuse, MCB, ELCB/RCD, Neutral link.

Q: Define power factor. Calculate the active power, reactive power, and apparent power for a load drawing 5A at 230V with power factor 0.8 lagging.

A: **Power Factor:** The ratio of real power to apparent power. PF = cos φ = P/S. **Given:** V = 230V, I = 5A, PF = 0.8. **S = V·I = 1150 VA. P = S·PF = 1150×0.8 = 920 W. Q = $\sqrt{S^2 - P^2}$ = √(1,322,500 − 846,400) = √476,100 ≈ 690 VAR.**

Module III — Static Electricity, ESD, Regulation

Q: What is Electrostatic Discharge (ESD)? Why is it a concern in electronics? List four ESD precautions.

A: **ESD** is the sudden flow of electricity between two objects at different potentials. It can damage sensitive electronic components (CMOS, MOSFETs, ICs) by causing catastrophic or latent failure. **Precautions:** (1) Use grounded wrist straps. (2) Use antistatic workstations and mats. (3) Store components in antistatic bags. (4) Maintain 40–60% relative humidity.

Q: Explain the working of a Zener diode as a voltage regulator. Draw the circuit.

A: A Zener diode is operated in the reverse breakdown region. When reverse-biased and the voltage exceeds V_Z, the diode conducts and maintains a nearly constant voltage. In the regulator circuit, a series resistor (R_s) limits current, and the Zener shunts excess current to maintain $V_{out} \approx V_Z$ across the load. The circuit diagram shows V_{in} → R_s → (Zener in parallel with load) → GND, with V_{out} taken across the Zener/load.

Q: Distinguish between the 78XX and 79XX series voltage regulators. What are the typical output voltages available?

A: **78XX:** Positive voltage regulators (e.g., 7805 = +5V, 7812 = +12V). **79XX:** Negative voltage regulators (e.g., 7905 = −5V, 7912 = −12V). Both are fixed-output IC regulators. Typical outputs: 7805 (+5V), 7809 (+9V), 7812 (+12V), 7815 (+15V), 7824 (+24V), and corresponding 79XX versions for negative voltages.

Module IV — Digital Electronics

Q: Convert $(1101\ 1010)_2$ to decimal and hexadecimal.

A: **To Decimal:** $(1101\ 1010)_2 = 1\times2^7 + 1\times2^6 + 0\times2^5 + 1\times2^4 + 1\times2^3 + 0\times2^2 + 1\times2^1 + 0\times2^0 = 128 + 64 + 0 + 16 + 8 + 0 + 2 + 0 = 218_{10}$. **To Hex:** Group into 4-bit nibbles: 1101 = D, 1010 = A → $(DA)_{16}$.

Q: Perform binary addition: $(1011)_2 + (0110)_2$. Also subtract $(0011)_2$ from $(1001)_2$ using 2's complement.

A: Addition: 1011 + 0110 = 10001 (with carry). Result = $(10001)_2 = (17)_{10}$.

Subtraction (1001 − 0011): 2's complement of 0011: 1100 + 1 = 1101. 1001 + 1101 = 10110 → discard carry → 0110. Result = $(0110)_2 = (6)_{10}$.

Q: Draw the symbol, write the truth table, and state the Boolean expression for NAND and ExOR gates.

A: NAND: Symbol = AND gate with bubble at output. Expression: $Y = \overline{(A \cdot B)}$. Truth table: 00→1, 01→1, 10→1, 11→0. **ExOR:** Symbol = OR-like with extra curve. Expression: $Y = A \oplus B = A\cdot\bar{B} + \bar{A}\cdot B$. Truth table: 00→0, 01→1, 10→1, 11→0.

PRACTICE

Model Question Paper

Based on the official pattern — this is a practice paper, not an official SITTTR model question paper.

Course: 2041 — Elements of Electrical & Electronics Engineering | **Time:** 2.5 hrs | **Marks:** 60

Part A — Very Short Answer (8 × 1 = 8 marks)

1. State Ohm's Law.
2. What is the SI unit of resistance?
3. What is the commercial unit of electrical energy?
4. Define power factor.
5. What is the output voltage of a 7812 IC?
6. What is the 1's complement of $(1010)_2$?
7. Name any two universal logic gates.
8. What is the function of an ELCB?

Part B — Short Answer (8 × 3 = 24 marks)

Answer any two out of three from each module (2×3×4 modules = 24 marks).

1. Explain Kirchhoff's Voltage Law with a numerical example. (M1)
2. Differentiate between AC and DC. (M1)
3. State and explain the Law of Resistance. (M1)
4. Explain the structure of a domestic electrical supply system. (M2)
5. What is earthing? Explain plate and pipe earthing. (M2)
6. Calculate the monthly electricity bill for a home with 4 bulbs (60W, 5 hrs/day), 1 fan (80W, 8 hrs/day). Tariff = ₹6/kWh. (M2)
7. What is ESD? List four ESD precautions. (M3)
8. Explain the working of a Zener diode as a voltage regulator. (M3)
9. Compare the 78XX and 79XX series regulator ICs. (M3)
10. Convert $(2A)_{16}$ to decimal and binary. (M4)
11. Explain binary subtraction using 2's complement with an example. (M4)
12. Draw the symbol and truth table for NAND and ExNOR gates. (M4)

Part C — Long Answer (4 × 7 = 28 marks)

One question from each module with internal choice.

1. **M1:** (a) State Kirchhoff's laws. Solve a circuit using KCL and KVL to find unknown currents and voltages. *OR* (b) Explain the generation of AC voltage. Represent a sinusoidal waveform and define all parameters.
2. **M2:** (a) Explain the types of wiring systems used in residential buildings. Compare conduit wiring with cleat wiring. *OR* (b) Define power factor and types of powers. A load draws 10A at 230V with PF 0.7. Calculate P, Q, S and the monthly energy if used for 8 hrs/day for 30 days.
3. **M3:** (a) Explain static electricity, ESD, and ESD precautions in detail. *OR* (b) Explain the need for voltage regulation. Describe the working of a Zener diode regulator and list the types of voltage regulator ICs.
4. **M4:** (a) Explain the conversion of numbers between binary, decimal, and hexadecimal systems with examples. *OR* (b) Explain binary arithmetic (addition, subtraction using 2's complement) and draw the truth table and logic symbols for all seven basic logic gates.

LAST-MINUTE

Quick Revision

📘 Module I — Laws & AC

- **Ohm's Law:** $V = IR$ (constant temp)
- **KCL:** $\Sigma I_{in} = \Sigma I_{out}$
- **KVL:** $\Sigma V = 0$ (loop)
- **Voltage Div:** $V_x = V_T \cdot R_x/R_T$
- **Current Div:** $I_x = I_T \cdot R_T/R_x$
- **Faraday:** $E = -N \cdot d\Phi/dt$
- **AC:** $v(t) = V_m \sin(\omega t + \varphi)$, $V_{rms} = V_m/\sqrt{2}$

🏠 Module II — Domestic Supply

- **Supply:** Single-phase (L+N, 230V), Three-phase (R+Y+B+N, 415V)
- **Protection:** Fuse, MCB, ELCB, Neutral link
- **Earthing:** Plate, Pipe — fault current path
- **Wiring:** Cleat, Batten, Casing-capping, Conduit (concealed preferred)
- **Powers:** $P = VI\cos\varphi$ (W), $Q = VI\sin\varphi$ (VAR), $S = VI$ (VA)
- **Energy:** $E = P \cdot t$ (kWh), 1 kWh = 3.6×10^6 J

⚡ Module III — ESD & Regulation

- **Static:** Charge buildup from friction
- **ESD:** Sudden discharge — damages electronics
- **Precautions:** Grounding, antistatic bags, humidity control
- **Zener regulator:** Reverse breakdown, $V_{out} \approx V_Z$
- **78XX:** Positive fixed (7805=+5V, 7812=+12V)
- **79XX:** Negative fixed (7905=−5V, 7912=−12V)

- **LM317:** Variable 1.25–37V

💻 Module IV — Digital

- **Number systems:** Binary (2), Decimal (10), Hex (16)
- **Conversions:** Group binary in 4s for hex
- **2's complement:** 1's + 1 — for subtraction
- **Logic gates:** NOT, AND, OR, NAND, NOR, ExOR, ExNOR
- **Universal gates:** NAND, NOR
- **Truth table:** All possible input combinations

⚠️ Common Mistakes to Avoid:

- Forgetting that Ohm's Law is only valid at constant temperature.
- Mixing up KCL (currents) and KVL (voltages) — remember KCL is for nodes, KVL for loops.
- Confusing active power (W) with apparent power (VA) — PF is the ratio.
- Forgetting the negative sign in Faraday's law (Lenz's law).
- Not checking the 2's complement carry bit in binary subtraction.
- Confusing NAND (AND + NOT) with NOR (OR + NOT).

✅ Exam Checklist — Before You Write:

- Read the question carefully — identify what is being asked.
- For numerical problems: write the formula, substitute values, show units.
- For circuit problems: draw a small schematic or label the diagram.
- For logic gates: draw the symbol, write the expression, show the truth table.
- For conversions: show the steps clearly.
- Manage time: Part A (1 mark) → short, Part B (3 marks) → medium, Part C (7 marks) → detailed.

TERMS

Glossary

Important Terms and One-line Meanings

Term	Meaning
Current	Rate of flow of electric charge (A)
Voltage	Electrical potential difference (V)
Resistance	Opposition to current flow (Ω)
Ohm's Law	V = IR (constant temperature)
KCL	Total current entering a node = total current leaving
KVL	Sum of voltages around a closed loop = 0
Faraday's Law	EMF induced by changing magnetic flux
AC	Alternating Current — reverses direction periodically
DC	Direct Current — flows in one direction
RMS	Root Mean Square — effective value of AC
MCB	Miniature Circuit Breaker — resettable overcurrent protection
ELCB	Earth Leakage Circuit Breaker — shock protection
Earthing	Connecting equipment to earth for safety
Power Factor	Ratio of real power to apparent power
kWh	Kilowatt-hour — commercial unit of energy
ESD	Electrostatic Discharge — sudden static discharge
Zener Diode	Diode used in reverse breakdown for voltage regulation
78XX	Positive fixed voltage regulator ICs
79XX	Negative fixed voltage regulator ICs
LM317	Adjustable positive voltage regulator
Binary	Base-2 number system (0,1)
Hexadecimal	Base-16 number system (0–9, A–F)
2's Complement	Method to represent negative binary numbers
NAND	Universal gate — AND followed by NOT
NOR	Universal gate — OR followed by NOT
ExOR	Exclusive OR — output 1 when inputs differ

RESOURCES

References

Textbooks

- **B. L. Theraja and A. K. Theraja** — *Textbook of Electrical Technology: Part 1 — Basic Electrical Engineering in S. I. Units.* S. Chand Publication.
- **J B Gupta** — *Electronic Devices and Circuits.* Katson Books.

References

- **N.N. Bhargava, D.C. Kulshreshtha, S.C. Gupta** — *Basic Electronics and Linear Circuits.* Tata McGraw Hill Education.
- **A. Anand Kumar** — *Fundamentals of Digital Circuits.* Prentice Hall India Pvt. Ltd.

Web Resources (Officially Listed)

- NPTEL — Digital Electronic Circuits (Course ID: 108105132)